

3 The sequence $u_1, u_2, u_3, \dots$ is such that $u_1 = 5$ and $u_{n+1} = 6u_n + 5$ for $n \geq 1$.

(a) Prove by induction that $u_n = 6^n - 1$ for all positive integers n . [5]

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(b) Deduce that u_{2^n} is divisible by u_n for $n \geq 1$. [2]

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